Q14: Block Diagram & Working of 8259 PIC [2022, 2023, 2024, 2025 – Very Important!]

8259 = Programmable Interrupt Controller. Manages up to 8 interrupt inputs (IR0-IR7) and can be cascaded for 64 interrupts.

Key Internal Blocks of 8259:

Block	Function
IRR (Interrupt Request Register)	8-bit register. Stores all pending interrupt requests (IR0-IR7). A bit is set when corresponding IR line goes HIGH.
IMR (Interrupt Mask Register)	8-bit register. Masks (disables) specific interrupts. Bit=1 → that interrupt is masked (disabled).
ISR (In-Service Register)	Tracks which interrupt is currently being serviced by the CPU.
Priority Resolver	Determines which pending interrupt has highest priority. Default: IR0 = highest, IR7 = lowest.
Data Bus Buffer	8-bit tri-state buffer connecting 8259 to system data bus.
Read/Write Control	Handles communication with CPU (ICW and OCW programming).
Cascade Buffer	Allows up to 8 slave 8259s to be connected to one master (for 64 total interrupts).

Interrupt Sequence (Step by Step):

Device raises interrupt on IR line (IR0–IR7). IRR bit is set.

1. Priority Resolver checks IMR – if not masked and highest priority, it sends INT signal to CPU.
2. CPU acknowledges with INTA pulse (first INTA: ISR bit set, IRR bit cleared).
3. Second INTA: 8259 places interrupt vector (type number) on data bus.
4. CPU jumps to ISR (Interrupt Service Routine). 8259 blocks lower-priority interrupts.
5. ISR issues EOI (End of Interrupt) command → ISR bit cleared → 8259 ready for next interrupt.

Control Word Bit Structure (Mode Definition Register):

Table

Examples:

```
; All ports as output, Mode 0: D7=1,D6D5=00,D4=0,D3=0,D2=0,D1=0,D0=0
Control Word = 80H  → All output, Mode 0

; Port A = Input, Port B = Output, Port C = Output, Mode 0
; D7=1,D6D5=00,D4=1,D3=0,D2=0,D1=0,D0=0 = 90H
Control Word = 90H

; Port A = Mode 1 Input, Port B = Mode 0 Output

; D7=1,D6D5=01,D4=1,D3=0,D2=0,D1=0,D0=0 = B0H
```

Bit	D7	D6 D5	D4	D3 D2	D1	D0
Field	Flag	Mode A (00/01/10)	Port A (0=Out, 1=In)	Mode B (00/01)	Port B (0=Out, 1=In)	Port C Lower
Value if D7=1	1 = I/O Mode	00=Mode0, 01=Mode1, 1X=Mode2	0=Output, 1=Input	00=Mode0, 01=Mode1	0=Output, 1=Input	0=Out, 1=In

Q. **Flag Status After Instruction Execution (ADD, ANI etc.)**

8085 Flags Overview:

Flag	Name	Set When
S	Sign	Result is negative (bit 7 = 1)
Z	Zero	Result is 00H (all zeros)
AC	Auxiliary Carry	Carry from bit 3 to bit 4
P	Parity	Number of 1-bits in result is even
CY	Carry	Result generates carry/borrow beyond 8 bits

Example 1: ADD Instruction

```
MVI A, 85H      ; A = 1000 0101
MVI B, 92H      ; B = 1001 0010
ADD B           ; A = A + B = 85H + 92H = 117H
```

Calculation: 85H + 92H = 0001 0001 0111 => Result = 17H, Carry = 1

Flag	Value	Reason
S	0	Bit 7 of result (17H = 00010111) = 0
Z	0	Result is 17H, not 00H
AC	1	Carry from lower nibble: 5+2=7, upper: 8+9+0=11 -> carry
P	1	17H = 00010111, count of 1s = 4 (even) -> Parity = 1
CY	1	85H + 92H = 117H, exceeds 8-bit limit

Example 2: ANI Instruction (AND Immediate)

```
MVI A, 6CH      ; A = 0110 1100
ANI 0FH         ; A = A AND 0FH = 0000 1100 = 0CH
```

Flag	Value	Reason
S	0	Bit 7 of 0CH (00001100) is 0
Z	0	Result is 0CH, not zero
AC	1	ANI always sets AC to 1
P	1	0CH = 00001100, two 1s (even), P = 1
CY	0	ANI always clears CY

Q6: ALP to Transfer Block of N Bytes (14 bytes from 4500H to 5500H)

```
; 16-bit addition: [3000H-3001H] + [3002H-3003H]
; Result stored at 3004H (low byte) and 3005H (high byte)
LXI H, 3000H   ; Load HL with address 3000H
MOV E, M       ; Load low byte of 1st number into E (from 3000H)
INX H          ; HL = 3001H
MOV D, M       ; Load high byte of 1st number into D (from 3001H)
INX H          ; HL = 3002H
MOV C, M       ; Load low byte of 2nd number into C (from 3002H)
INX H          ; HL = 3003H
MOV B, M       ; Load high byte of 2nd number into B (from 3003H)
MOV H, D       ; Move 1st number high byte to H
MOV L, E       ; Move 1st number low byte to L  => HL = 1st 16-bit no.
MOV D, B       ; DE = 2nd 16-bit number
DAD D          ; HL = HL + DE (16-bit addition)
LXI B, 3004H   ; Store result
MOV M, L       ; Store low byte at 3004H
INX B          ; Point to 3005H
MOV M, H       ; Store high byte at 3005H
HLT            ; Halt
```

Explanation:
LXI H: Loads 16-bit immediate address into HL register pair.
MOV D/E from M: Picks up memory contents byte by byte.
DAD D: Adds contents of DE to HL, result in HL. Carry flag set if overflow.
Result low byte at 3004H, high byte at 3005H.
DAD instruction affects only the Carry flag (CY).

Q6: ALP to Transfer Block of N Bytes (14 bytes from 4500H to 5500H)

```
; Block Transfer: N bytes from source (4500H) to destination (5500H)
LXI H, 4500H   ; HL = Source address
LXI D, 5500H   ; DE = Destination address
MVI C, 0EH     ; C = 14 (0EH) = byte count
LOOP:
MOV A, M       ; Load byte from source (address in HL) into A
STAX D         ; Store byte to destination (address in DE)
INX H          ; Increment source pointer
INX D          ; Increment destination pointer
DCR C          ; Decrement counter
JNZ LOOP       ; If C != 0, repeat
HLT            ; All 14 bytes transferred
```

Explain Pins of 8085: ALE, HOLD, INTR, READY, CLK OUT

Pin	Type	Description
ALE (Address Latch Enable)	Output (Active High)	ALE goes HIGH during the first clock cycle of every machine cycle. It signals external circuitry that the AD0-AD7 lines carry a valid ADDRESS (not data). Used to latch the low-order address byte using a latch IC like 74LS373.
HOLD	Input (Active High)	When HOLD=1, an external device (like DMA controller) requests to take control of the bus. The 8085 finishes the current machine cycle, then tri-states all buses (AD0-AD7, A8-A15, control lines) and issues HLDA.
HLDA (Hold Acknowledge)	Output	Sent in response to HOLD. When HLDA=1, the buses are released to the requesting device.
INTR (Interrupt Request)	Input (Active High)	A maskable interrupt input. If INTR=1 and interrupts are enabled (EI), 8085 completes current instruction and acknowledges via INTA. Lowest priority among hardware interrupts.
READY	Input (Active High)	Used by slow memory/IO devices. If READY=0, the 8085 inserts wait states (WAIT states) to extend the machine cycle until the device is ready. When READY=1, normal operation resumes.
CLK OUT (Clock Output)	Output	The 8085 outputs its internal clock on this pin at half the crystal frequency. Used to drive external devices that need the CPU clock signal. If 6 MHz crystal is used, CLK OUT = 3 MHz.

```
; Block Transfer: N bytes from source (4500H) to destination (5500H)
LXI H, 4500H   ; HL = Source address
LXI D, 5500H   ; DE = Destination address
MVI C, 0EH     ; C = 14 (0EH) = byte count
LOOP:
MOV A, M       ; Load byte from source (address in HL) into A
STAX D         ; Store byte to destination (address in DE)
INX H          ; Increment source pointer
INX D          ; Increment destination pointer
DCR C          ; Decrement counter
JNZ LOOP       ; If C != 0, repeat
HLT            ; All 14 bytes transferred
```

Step-by-Step Trace (first 2 iterations):

Iteration	A (Data)	HL (Src)	DE (Dst)	C (Count)
1	[4500H]	4500H → 4501H	5500H → 5501H	14 → 13
2	[4501H]	4501H → 4502H	5501H → 5502H	13 → 12
...	...	...	...	...
14	[450DH]	450DH → 450EH	550DH → 550EH	1 → 0 (HLT)